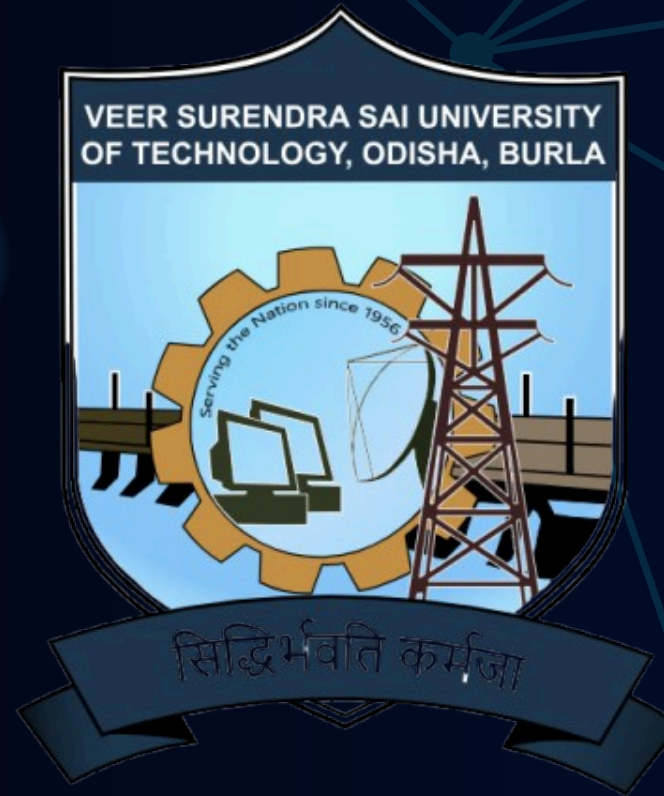


AN AI-POWERED NATURAL LANGUAGE INTERFACE FOR DATASET EXPLORATION



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PRESENTED BY:

SAHIL SAHOO

REGD.NO.:2302041048

OUTLINES:

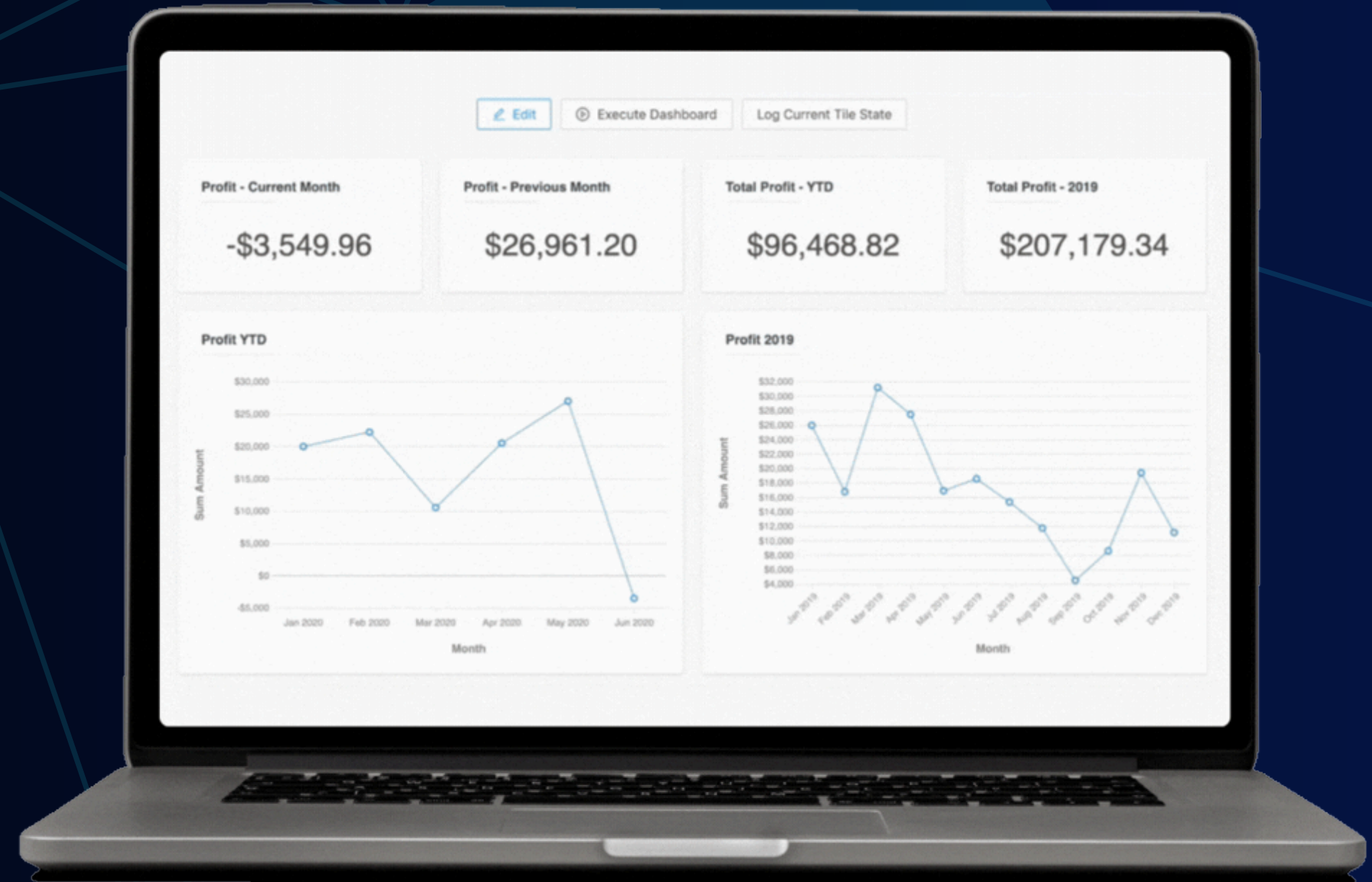
- ➔ INTRODUCTION
- ➔ OBJECTIVES
- ➔ WORKING PRINCIPLE
- ➔ TECHNOLOGIES

- ➔ ADVANTAGES
- ➔ CHALLENGES
- ➔ FUTURE VISION
- ➔ CONCLUSION

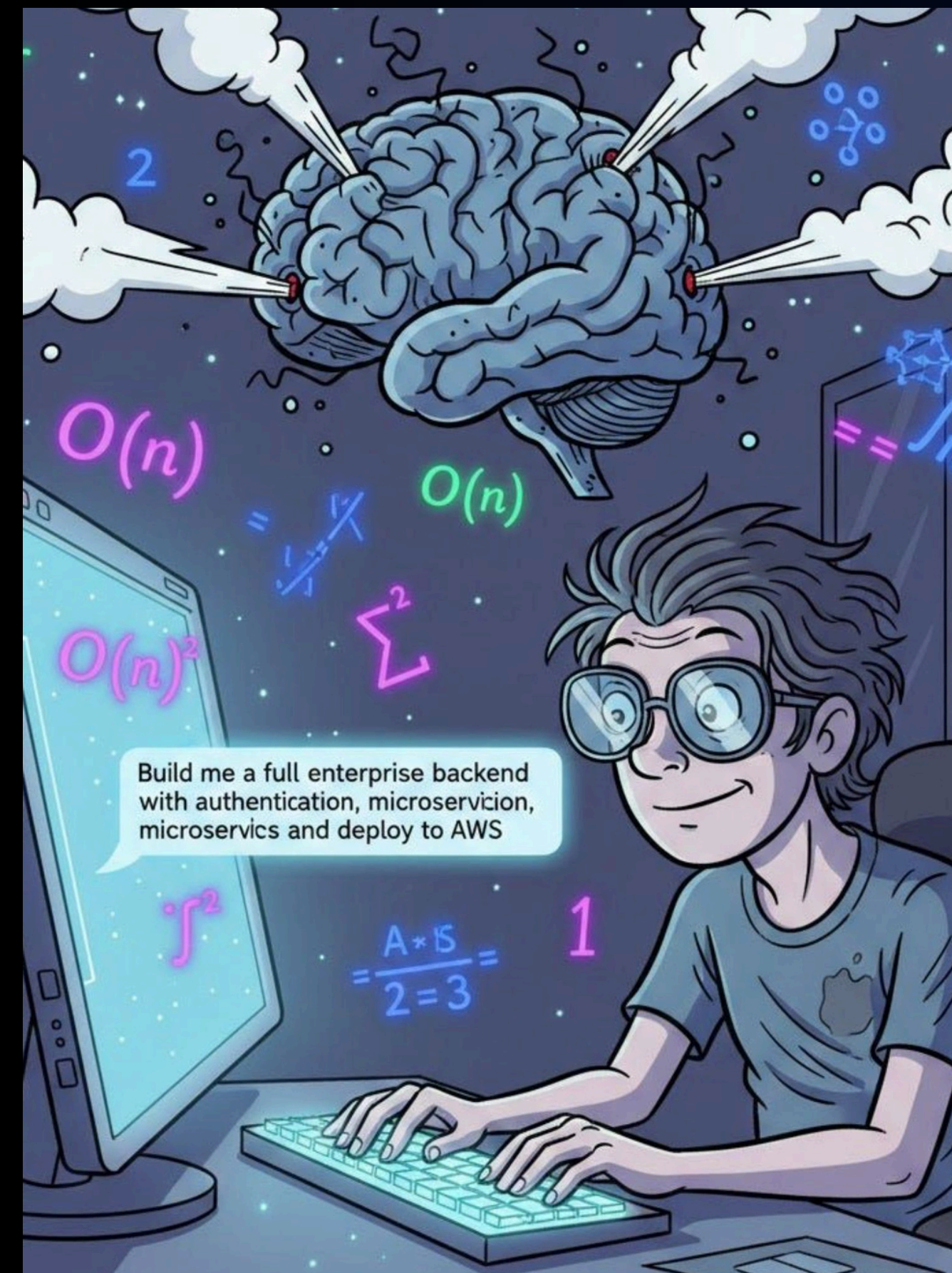
INTRODUCTION

This system is an AI-powered intelligent interface that allows users to interact with structured datasets using natural language. Instead of writing Python or SQL commands, users simply ask questions like:

The system interprets the query, generates executable code, processes the dataset, and returns a result, explanation, and visualization — all automatically.



THE PROBLEM WE'RE SOLVING



➔ Traditional data analysis tools demand technical proficiency that most non-programmers simply don't have. This creates a significant barrier between users and their own data.

Traditional Path	The Gap
Learn Python / SQL → Write code → Clean data → Perform operations → Create charts → Interpret results	Existing AI chatbots understand language but are not connected to the user's dataset or analytical workflow.
Our Approach	
Ask a question → AI understands intent → Generates code → Executes → Returns insight + visualization	

OBJECTIVES

The core goal is to develop an **AI-driven natural-language interface** for intelligent and automated exploration of structured datasets — reducing manual effort and expanding accessibility.



Easy Dataset Upload

Upload CSV or structured datasets with minimal configuration.



Automatic Code Generation

Generate executable Python/Pandas operations without user intervention.



Interactive Web Interface

A clean, conversational UI accessible to technical and non-technical users.



Natural Language Understanding

Interpret plain-English questions and automatically identify user intent.



Automated Visualization

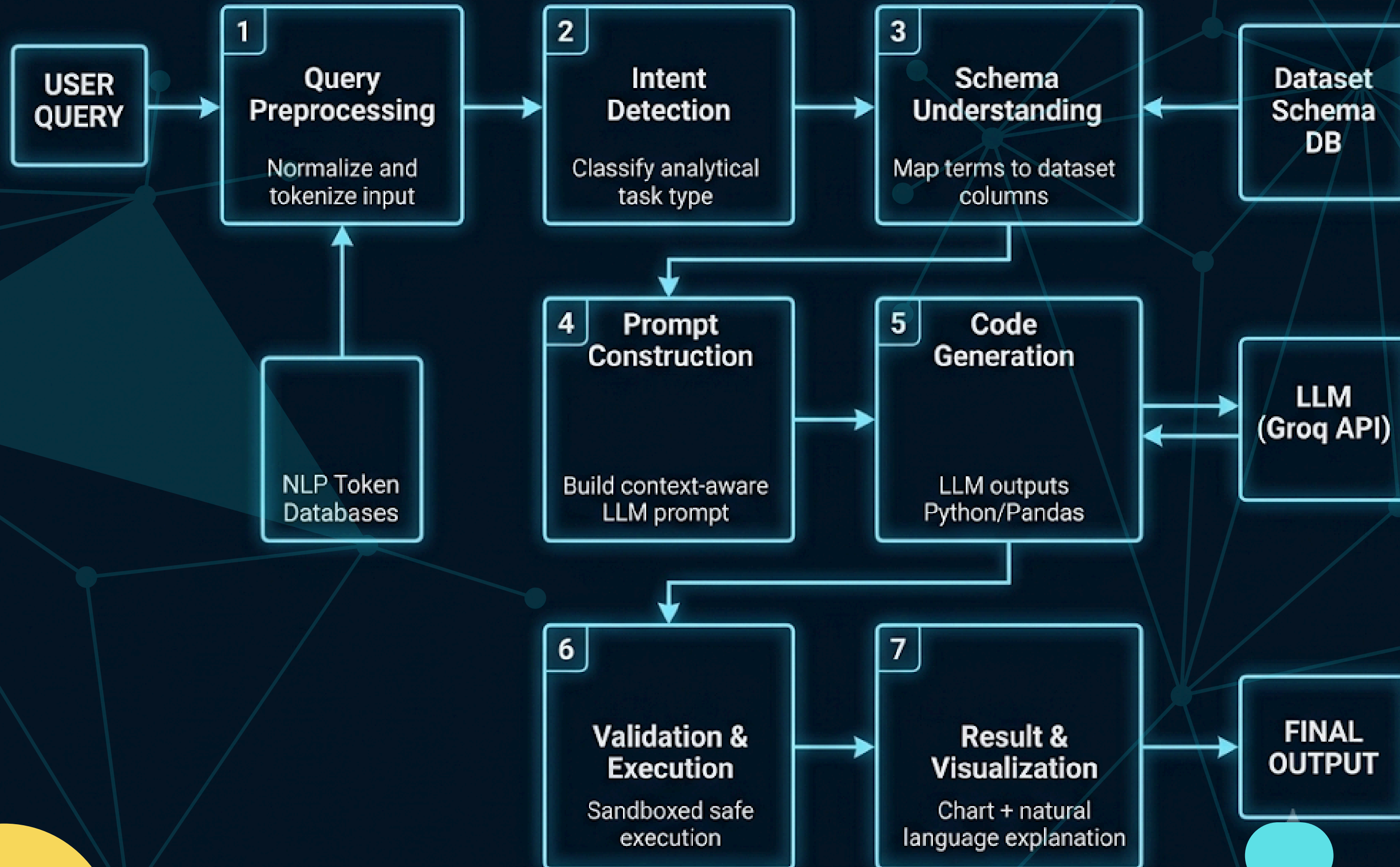
AI selects and renders the most appropriate chart for each query.



AI-Generated Insights

The system explains results in natural language, not just raw numbers.

WORKING PRINCIPLE



WORKING PRINCIPLE



STAGE 01 · FRONTEND

Web Interface

USER INPUT

🔍 Show revenue trend by region

• REQUEST SENT

MECHANISM

- Browser renders the HTML / CSS UI
- Flask receives the POST request
- Session state is attached & routed onward

START

INPUT

UNDERSTAND

GENERATE

EXECUTE

DELIVER

DONE

TECHNOLOGIES POWERING THE SYSTEM



Large Language Models

Understand natural-language queries and generate precise analytical instructions.



Generative AI

Automatically produces Python operations, explanations, and chart selections.



Natural Language Processing

Converts human language into structured analytical intent.



Groq API

High-speed LLM inference for real-time query processing.



Pandas + Python

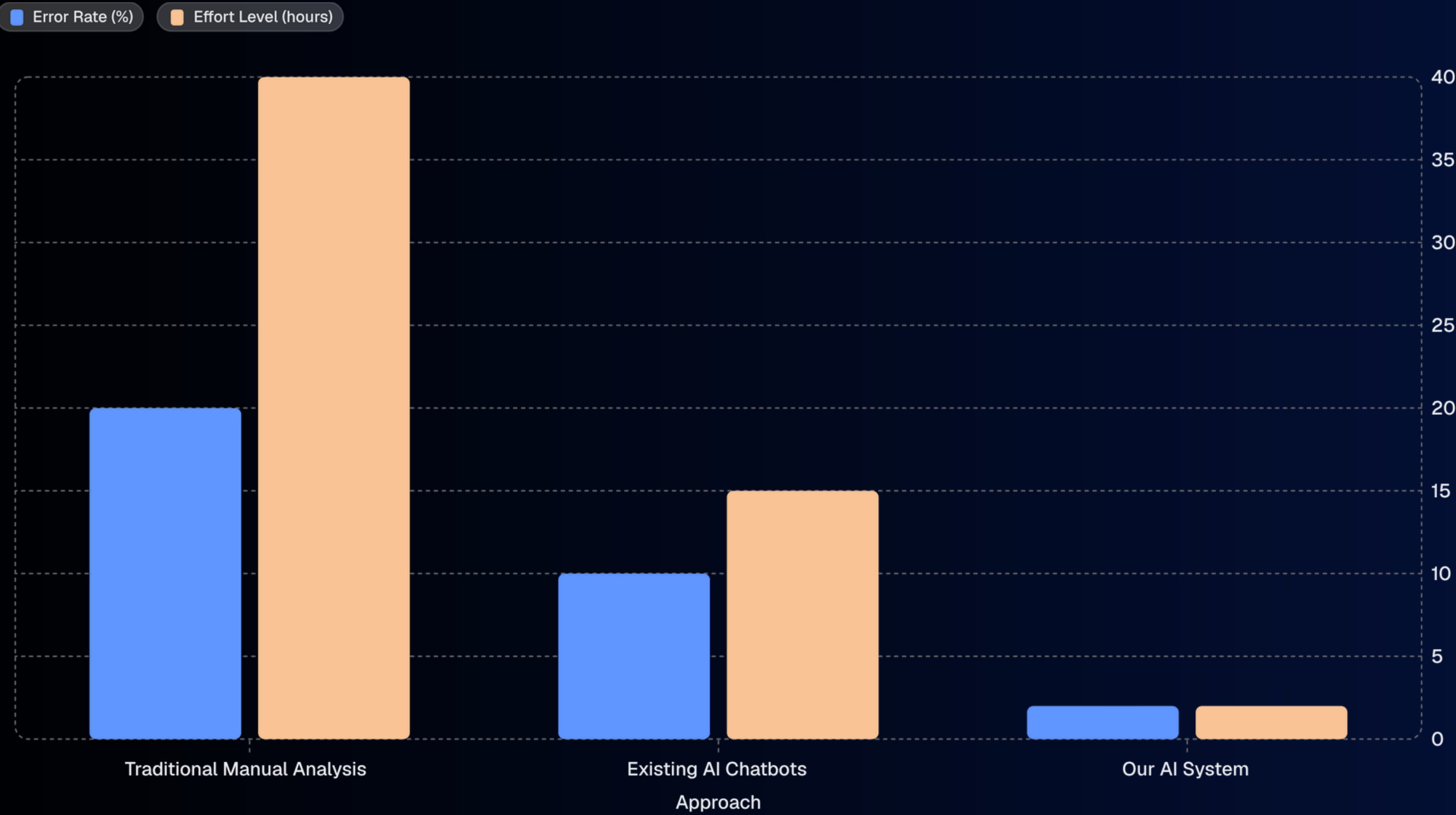
Core data processing engine for filtering, aggregation, and statistics.



Flask Backend

Lightweight Python web framework serving the interactive interface.

ERROR RATE & EFFORT COMPARISON

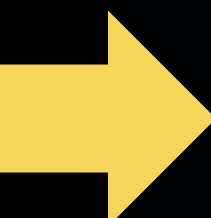


This comparison highlights how our AI System significantly reduces both the error rate and the human effort required for data exploration, drastically outperforming traditional manual analysis and existing AI chatbot solutions.

ADVANTAGES

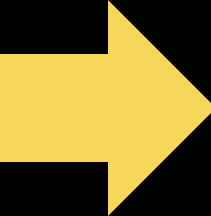
- ➔ **Natural Interaction** — Communicate with data in plain English
- ➔ **Reduced Coding** — No manual Python or SQL required
- ➔ **Automation** — Multiple analytical steps handled automatically
- ➔ **Faster Exploration** — Interactive, conversational querying
- ➔ **Accessibility** — Designed for non-technical users
- ➔ **AI-Generated Insights** — Explanations, not just numbers

CURRENT CHALLENGES



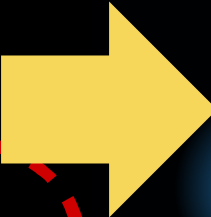
LLM Accuracy

Ambiguous queries may be misinterpreted.



Code Security

AI-generated code requires sandboxed execution.



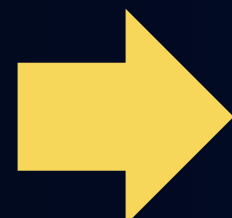
Context Memory

Follow-up questions require conversational state.



Hallucination Risk

LLMs may generate incorrect operations.



Large Datasets

Large files may increase processing time.

FUTURE VISION

1

AI Agent Architecture

Specialized tools for Python, SQL, and visualization orchestrated by an intelligent agent.

2

RAG & Semantic Layer

Domain-specific knowledge and metadata to improve response accuracy.

3

Multimodal & Voice

Voice queries, multilingual support, and AI-generated dashboards.

4

Predictive AI

Anomaly detection, forecasting, and cloud-based collaborative analysis.

"Our goal is to make interaction with data as simple as having a conversation."

CONCLUSION:

The project demonstrates how Generative AI and Natural Language Processing can simplify interaction with structured datasets. Users can ask questions in natural language, while the system interprets the query, generates the required operations, processes the dataset, and presents meaningful insights and visualizations.

Overall, the system reduces manual effort, improves accessibility, and makes intelligent dataset exploration faster and more user-friendly.



REFERENCES:

- [1] K. Affolter, K. Stockinger, and A. Bernstein. “A Comparative Survey of Recent Natural Language Interfaces for Databases.” The VLDB Journal, 28:793–819, 2019.
- [2] R. C. A. Iacob, F. Brad, E.-S. Apostol, C.-O. Truică, I. A. Hosu, and T. Rebedea. “Neural Approaches for Natural Language Interfaces to Databases: A Survey.” Proceedings of the 28th International Conference on Computational Linguistics (COLING), pp. 381–395, 2020.
- [3] N. Deng, Y. Chen, and Y. Zhang. “Recent Advances in Text-to-SQL: A Survey of What We Have and What We Expect.” Proceedings of the 29th International Conference on Computational Linguistics (COLING), pp. 2166–2187, 2022.
- [4] G. Katsogiannis-Meimarakis and G. Koutrika. “A Survey on Deep Learning Approaches for Text-to-SQL.” The VLDB Journal, 32:905–936, 2023.
- [5] M. Liu and J. Xu. “NLI4DB: A Systematic Review of Natural Language Interfaces for Databases.” arXiv preprint arXiv:2503.02435, 2026.



THANK YOU

For the attention

